Red-black trees exercises

13-E01 Black-height and height conventions

Root20B has children10R,30R and otherwise NIL. List bh for the three internal nodes and NIL, h, and each complete black count including root. Recompute with both children black. Prove the red/black child bh relations and derive the height bound from $n \geq 2^{bh(root)} - 1$ and nonadjacent reds. Do not reverse “at least half black” into “at most half black.”

13-E02 Old parent and beta pointers

p30.left=x10, x.right=y20, and y.left=β15. Other subtrees satisfy BST order. List every relevant parent/child field after LEFT-ROTATE(x), then give the p=NIL root branch. Why is “y's original parent points to x” the wrong external wiring? Why must rotation avoid arbitrary shared NIL.parent writes when β=NIL?

13-E03 Two rotations for inner insertion

Insert35,15,25 into an empty tree. Draw nodes/colors after the last insertion, after I2, and after I3. Track z and each rotation center. Prove I2/I3 black-count conservation with four black-rooted boundary subtrees, rather than checking only three keys.

13-E04 Red uncle: complete counts versus bh

Insert25,15,35,5. List g,p,u,z and their colors before I1, after I1, and after final root blackening. Compute complete black counts including g and bh(g) excluding it. If g is not root, does every whole-tree path gain one black? State the only possible violation after moving repair upward.

13-E05 Preserve data while moving a successor

Insert40,20,60,10,30,50,70,45,55 and delete40. Identify z,y,x, y's original color, NIL.parent, and each TRANSPLANT. Draw the final tree. Does y's new black root color require black repair? Which of original data D45 and D40 remains?

13-E06 Red sibling followed by red-parent absorption

Insert15,25,35,45,55,65 and delete15. Draw complete trees after splice, after D1, and after D2 plus final blackening. Mark x's empty position, parent, and w. Using b, prove D1 preserves effective counts. Why cannot D2 after D1 continue up a black-parent chain?

Red-black trees exercises (continued)

13-E07 Near-red, far-black must proceed to D4

Insert15,25,45,35 and delete15. Give root,p,w,near,far and colors before/after D3 and after D4. Draw extra's position. Using p's original black contribution c and x's actual count b, prove D4 preserves the original effective local count. May D3 terminate directly?

13-E08 Black-parent D2 and deletion of the last node

Start with25B and black leaves15B,35B; delete15. Draw D2 and root absorption and compute root bh before/after. Then delete singleton black root25, stating root,x,NIL.parent, and final color. Does the empty tree retain all five properties?

13-E09 Right deficit and direct successor

Insert15,35,45,25 and delete35. Why must y.parent=z lead to x.parent=y? Give right-side D3/D4 rotation directions, final tree, and undeleted record identities. If only sibling changes to left in left-side pseudocode, without mirroring near/far and rotations, which premise fails?

13-E10 Height bounds versus rotation bounds

What bounds on n and h follow from black height3? What is the general height upper bound at n=1023? Distinguish complete traversal from one search. Prove at most2 insertion and3 deletion rotations while total ancestor work may be Θ(log n). Explain specifically why D1 cannot occur twice.

13-E11 Sentinel and API invariants

Which should be checked in a stable tree: (a) correct non-NIL child.parent; (b) shared NIL.parent equals the parent of every empty child position; (c) NIL black; (d) root.parent=NIL; (e) global key order and all NIL-path black counts? Specify duplicate insertion returning the original record and deletion of an absent key. Why cannot NIL be deleted as a normal keyed node?

13-E12 Rotation alone does not preserve red-black properties

Start with20B and children10R,30R. Left-rotate20 without recoloring. List inorder, root color, and black counts from the new root to every NIL type. Which properties fail? Does blackening only the new root repair them? Why does this not contradict rotation preserving BST order?

Red-black trees exercises (continued)

13-E13 Contract red links and test the assumptions

Difficulty: advanced

1. Black root28 has red children12,44. Node12 has black children5,20; node44 has black children36,55. All other children are black NIL. Contract red links, list the keys and four boundary intervals in the2-3-4 tree, and compute bh(28) and the complete black count including the root.
2. Do BST order, a black root, and equal root-NIL black counts alone guarantee at most three keys per contracted node? Give a rightward red-chain counterexample and identify the missing red-black property.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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